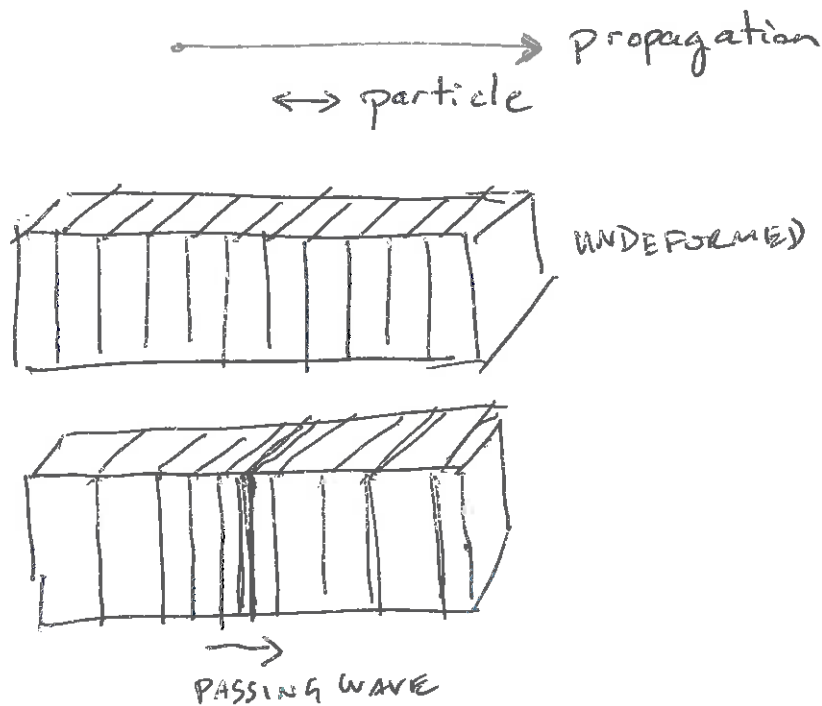


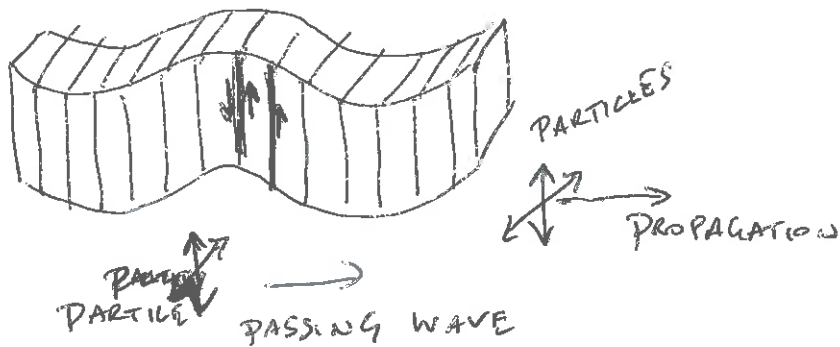
## P - WAVES

COMPRESSIONAL / DILATIONAL

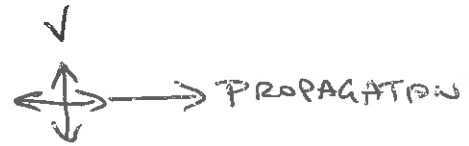
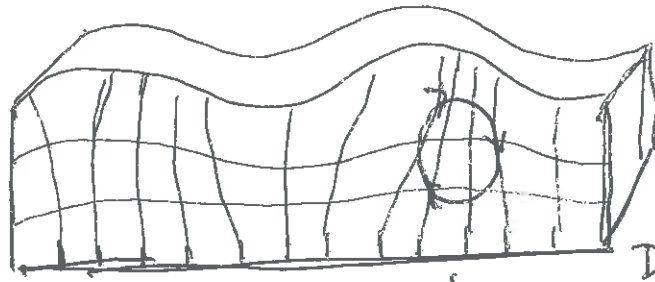


## S - WAVES

SHEAR WAVES



|           |         |                 |
|-----------|---------|-----------------|
| P-WAVES , | S-WAVES | (BODY WAVES)    |
| RALEIGH   | LOVE    | (SURFACE WAVES) |



DEPTH, NO PARTICLE MOTION

RAYLEIGH (LR)

SV - WAVE, P - WAVE TRAPPED AT SURFACE

WAVELENGTH  $\lambda \rightarrow$  INC,  $z \rightarrow$  INC

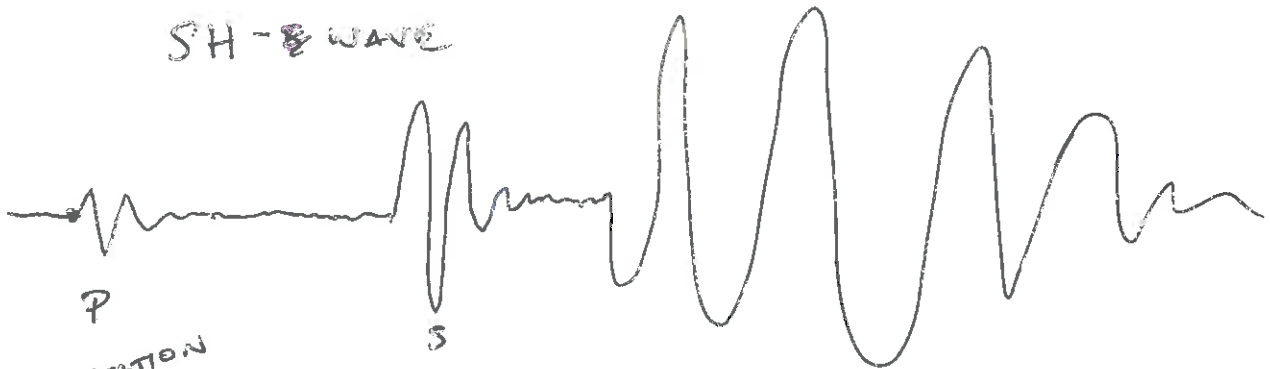
FREQUENCY  $\omega \leftarrow$  DEC,  $z \rightarrow$  INC

LOVE WAVE (LQ)

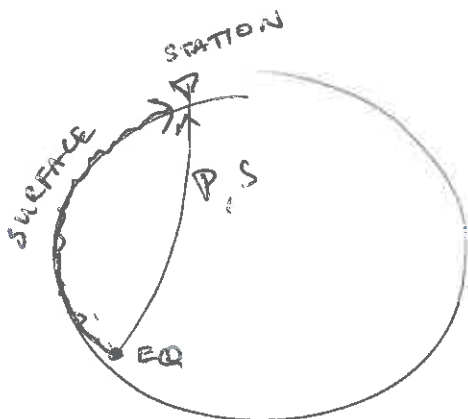


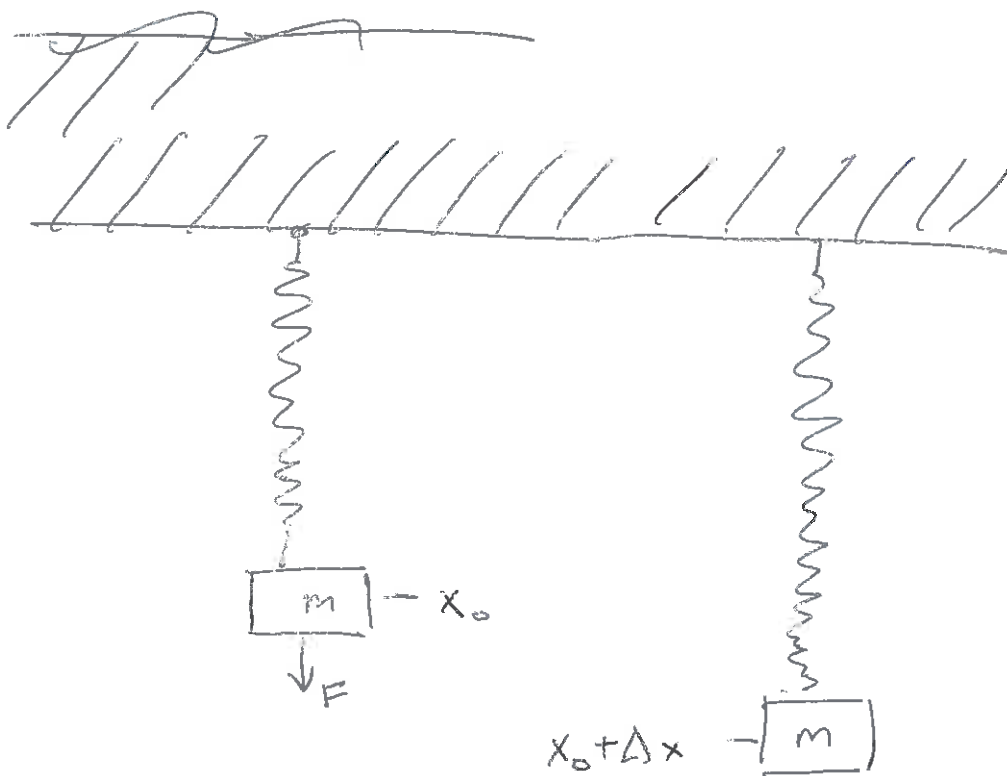
PROPAGATION

SH - WAVE



SURFACE WAVES





$$F = k \Delta x \quad \text{HOOKE'S LAW}$$

$$\text{STRESS} \left( \frac{F}{A} \right) = \left[ \frac{k x_0}{A} \right] \left( \frac{\Delta x}{x_0} \right) \text{STRAIN}$$

NEW CONSTANT

$$\sigma = E \epsilon$$

$$\vec{\sigma}_{ij} = \lim_{A_j \rightarrow 0} \frac{\vec{F}_i}{A_j}$$

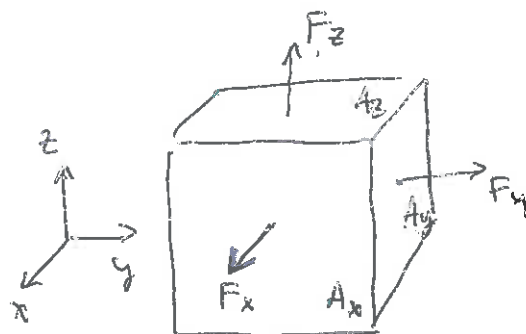
$i=j$  NORMAL STRESS

$$\vec{\sigma}_{xx} = \lim_{A_x \rightarrow 0} \frac{\vec{F}_x}{A_x}$$

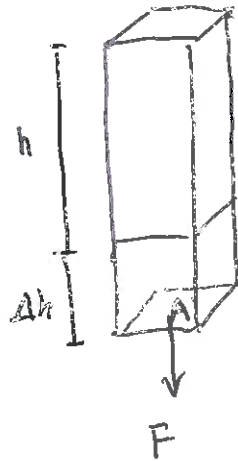
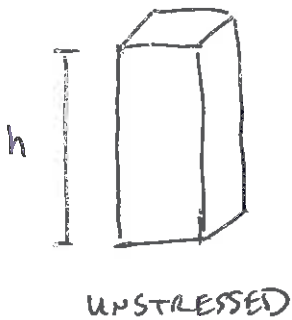
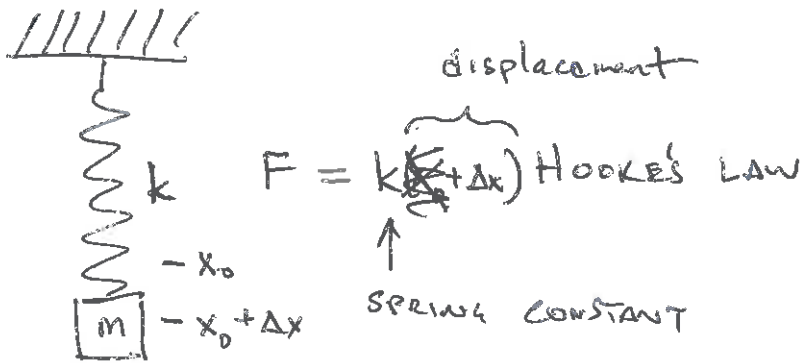
$$\vec{\sigma}_{yy} = \lim_{A_y \rightarrow 0} \frac{\vec{F}_y}{A_y}$$

$$\vec{\sigma}_{zz} = \lim_{A_z \rightarrow 0} \frac{\vec{F}_z}{A_z}$$

STRESS



3



$$\epsilon = \frac{\Delta h}{h}$$

$$\frac{\text{Force}}{\text{Area}} = \sigma, \text{ STRESS}$$

$$\frac{\left[ \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \right]}{\left[ \text{m}^2 \right]} = \frac{\left[ \text{N} \right]}{\left[ \text{m}^2 \right]} = \left[ \text{Pa} \right], \text{ pascals}$$

$$\sigma_{ij} = \lim_{A_j \rightarrow 0} \frac{F_i}{A_j}$$

$$i=j$$

NORMAL STRESS

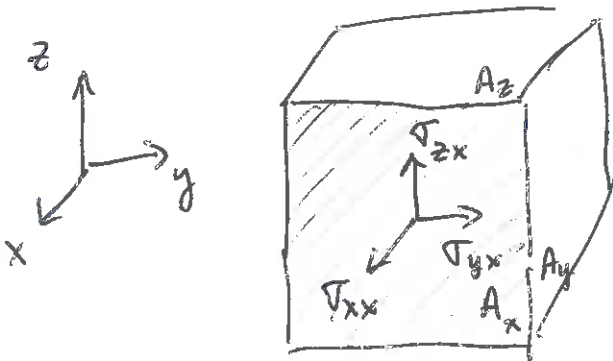
$$\sigma_{xx} = \lim_{A_x \rightarrow 0} \frac{F_x}{A_x}$$

$$\sigma_{yy} = \lim_{A_y \rightarrow 0} \frac{F_y}{A_y}$$

$$\sigma_{zz} = \lim_{A_z \rightarrow 0} \frac{F_z}{A_z}$$

(i) SHEAR STRESS

$$\tau_{xy} = \lim_{A_y \rightarrow 0} \frac{F_x}{A_y}$$



u DISPLACEMENT FIELD

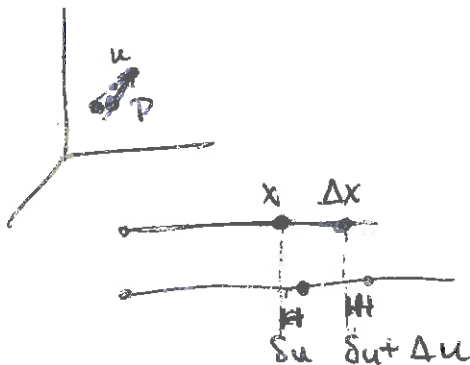
$$u = u(x, y, z)$$

NORMAL STRAINS

$$\epsilon_{xx} = \frac{\partial u_x}{\partial x}$$

$$\epsilon_{yy} = \frac{\partial u_y}{\partial y}$$

$$\epsilon_{zz} = \frac{\partial u_z}{\partial z}$$



$$\frac{\Delta V}{V} = \Delta = \epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}$$

DILATION

$$\Delta = \nabla \cdot u$$

~~DE~~ SCALAR DISPLACEMENT POTENTIAL

WAVE EQUATION

$$\frac{\partial^2 \Delta}{\partial t^2} = \alpha^2 \nabla^2 \Delta$$

P-WAVES

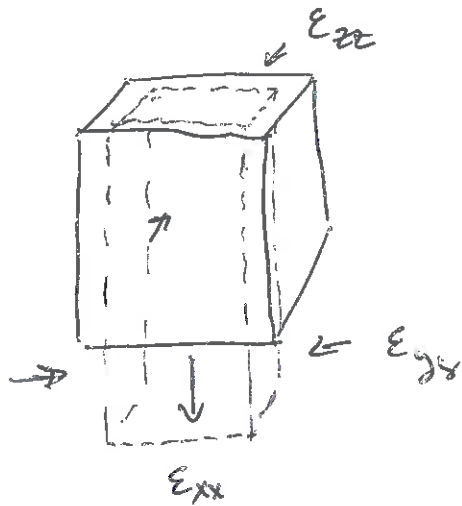
$$\frac{\partial^2 \Delta}{\partial t^2} = \alpha^2 \nabla^2 \Delta$$

P-WAVE VELOCITY

(5)

$$\alpha = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}}$$

K - Bulk modulus

 $\mu$  - Shear modulus $\rho$  - Density

$$\epsilon_{xx} = -\nu \epsilon_{yy}$$

$$\epsilon_{xx} = -\nu \epsilon_{zz}$$

 $\nu$  = Poisson's Ratio

$$\sigma_{xx} = E \epsilon_{xx}$$

E = Young's modulus.

$$E = \frac{\mu}{2(1+\nu)}, \quad E = \frac{3K(1-2\nu)}{2}$$

 $\nu$   
 $\mu$